

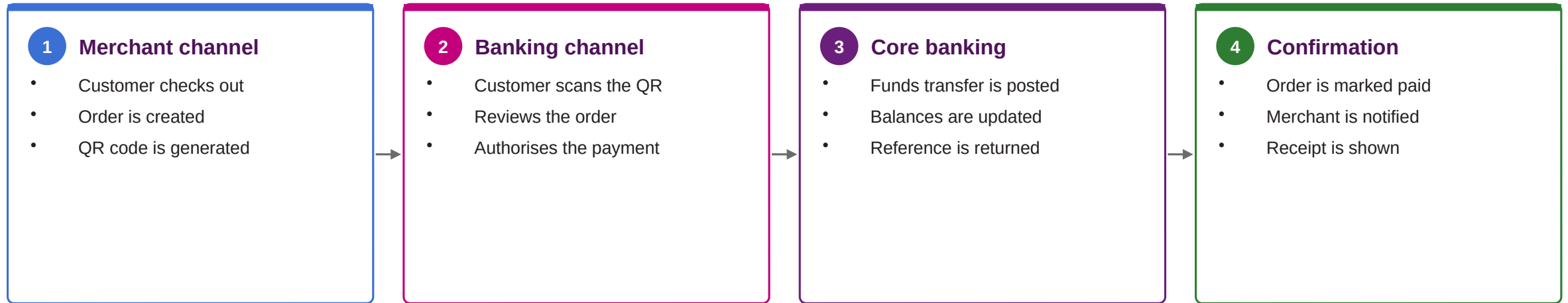
QR Payment Platform

Solution Design

Presented by TechStart

Merchant QR generation · Bank-side scanning and authorisation · Posting in the core banking system

Solution overview



Design principle

The merchant channel never holds funds and never moves money. It generates a QR that references an order. Every debit, credit and balance stays inside the core banking system.

QR is single-use

Time-bound — 300 s

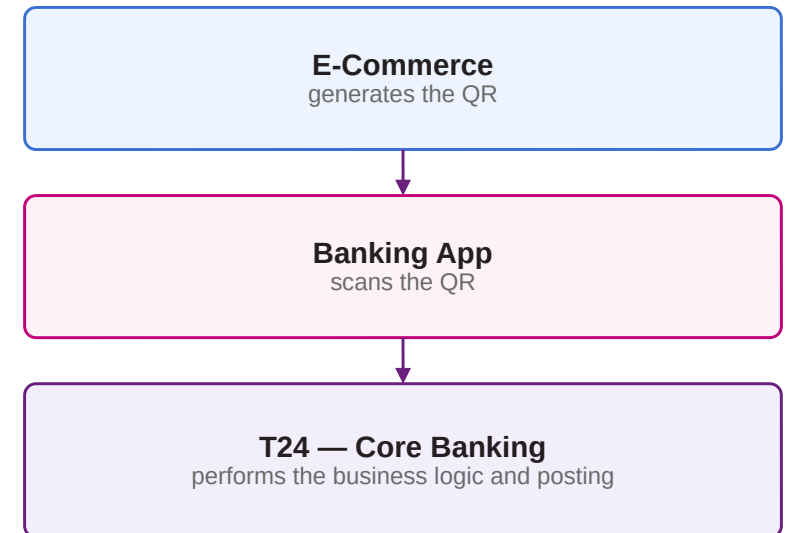
Customer authenticates first

Posting happens in T24

Business requirements

- 1 **E-Commerce generates the QR** at checkout, from the order
- 2 **Banking app scans the QR** to initiate the payment
- 3 **Core banking performs the posting** debit payer, credit merchant
- 4 **QR is single-use and time-bound** expires after 300 seconds
- 5 **Customer authenticates before authorising**
- 6 **Merchant is notified on confirmation**
- 7 **Order and payment are traceable end to end**

The core requirement



The bank remains the system of record.

Assumptions

#	Assumption
1	There are no budget constraints on the solution.
2	A third-party payment gateway can provide this capability. This design assumes an in-house build using the bank's own channels and core system.
3	E-Commerce is the QR generator only. It never holds funds and never moves money.
4	The banking app is the scanning and authorisation channel only.
5	All business logic, posting and balances remain in the core banking system.
6	The core banking system is T24.
7	The existing microservice and API management layer can be reused for the integration.
8	Volumes are pilot scale. A production throughput target will be set separately.

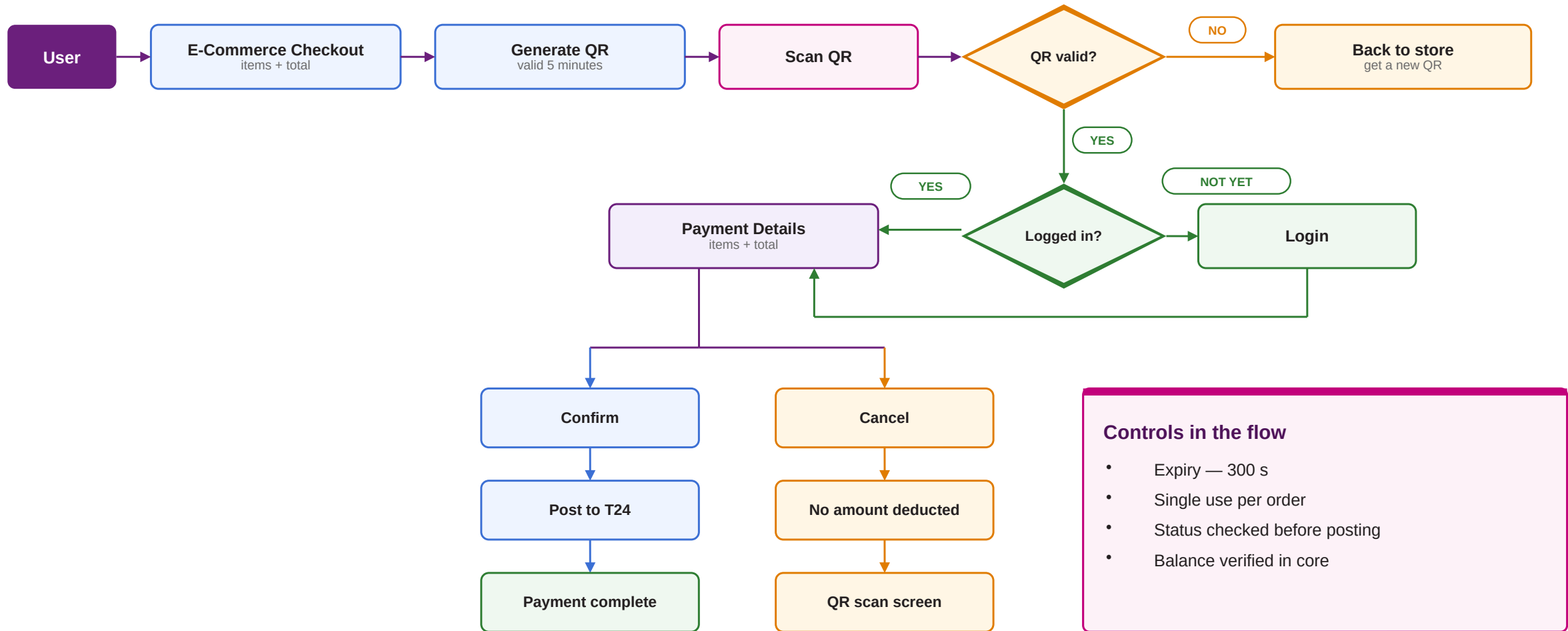
Build vs buy

- A gateway would shorten delivery
- An in-house build keeps the ledger and the customer journey inside the bank

Boundary

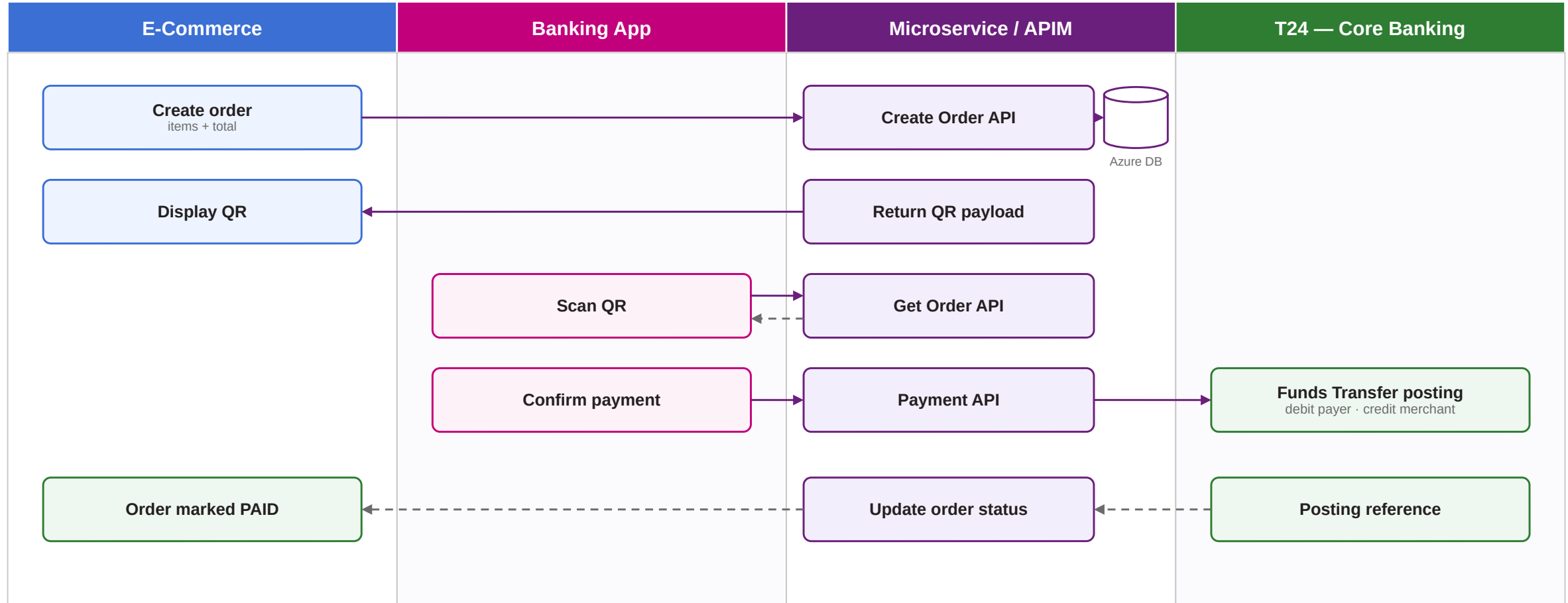
- Channels present and capture
- T24 decides and records
- Nothing settles outside the core

User flow



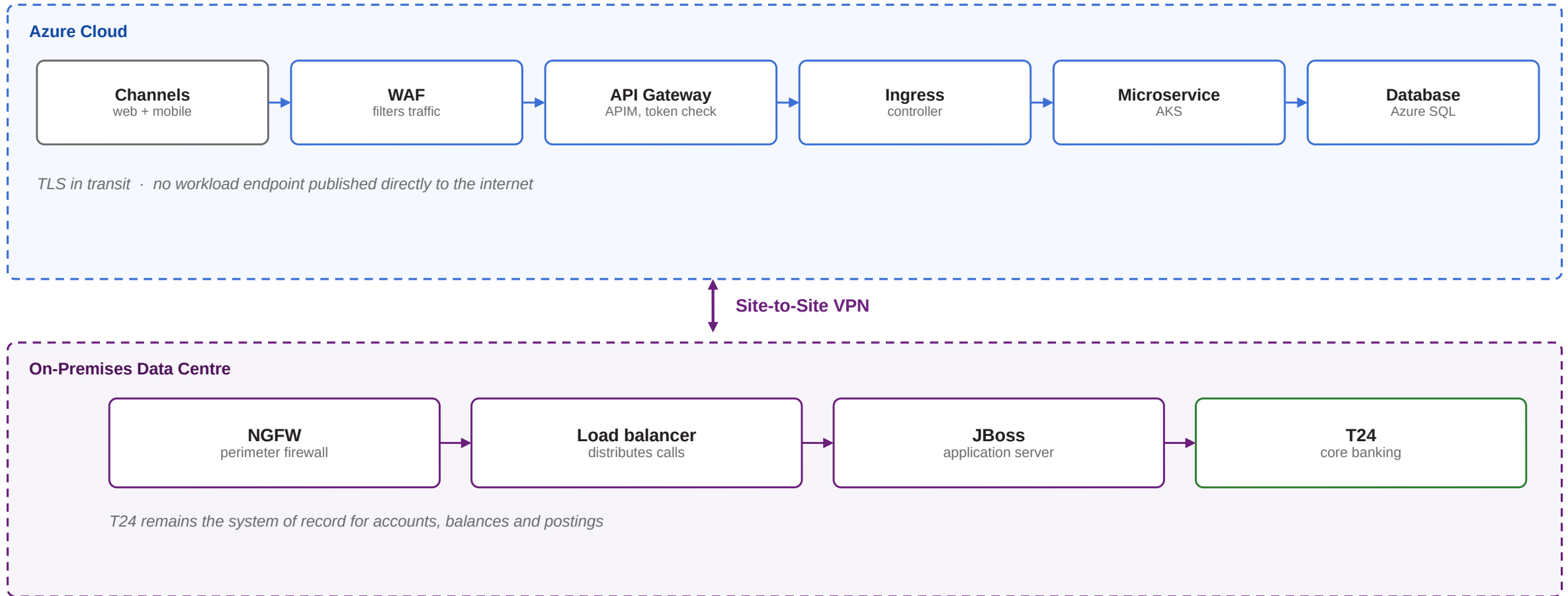
Solution architecture

Channels present and capture. The core banking system decides and records.



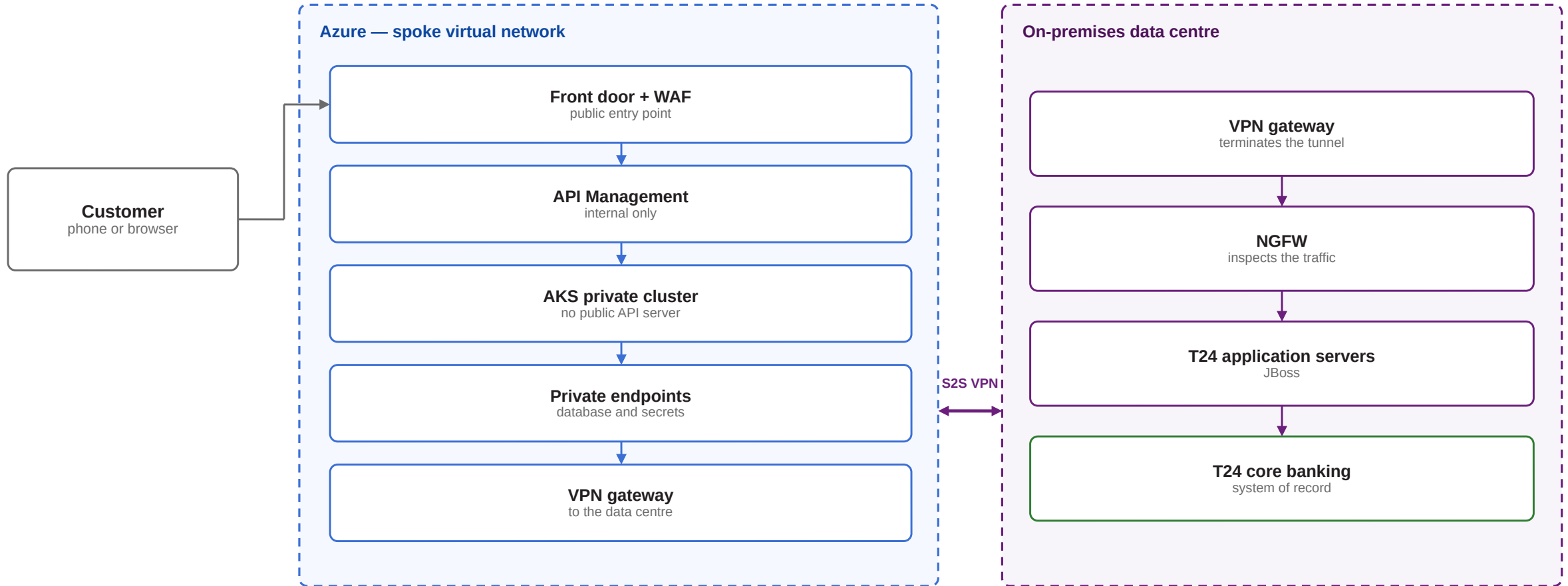
System interfaces

Channels reach the core banking system through API management and a site-to-site link.



Network and remote access

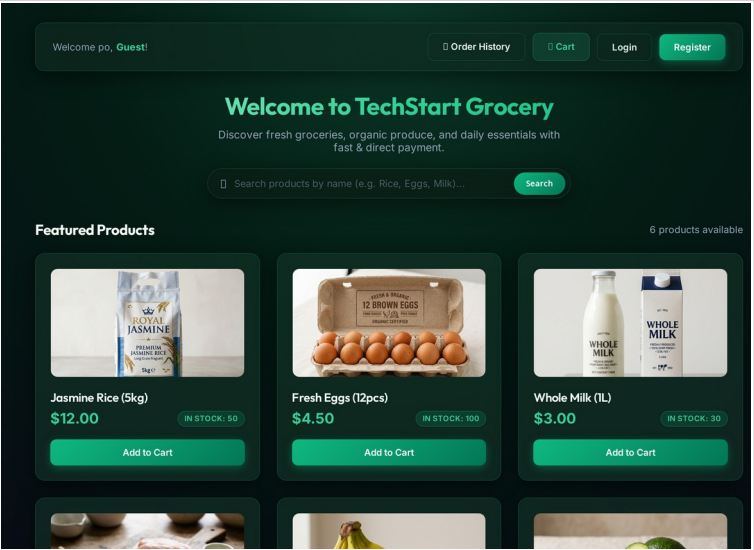
Private connectivity between the Azure workload and the on-premises core.



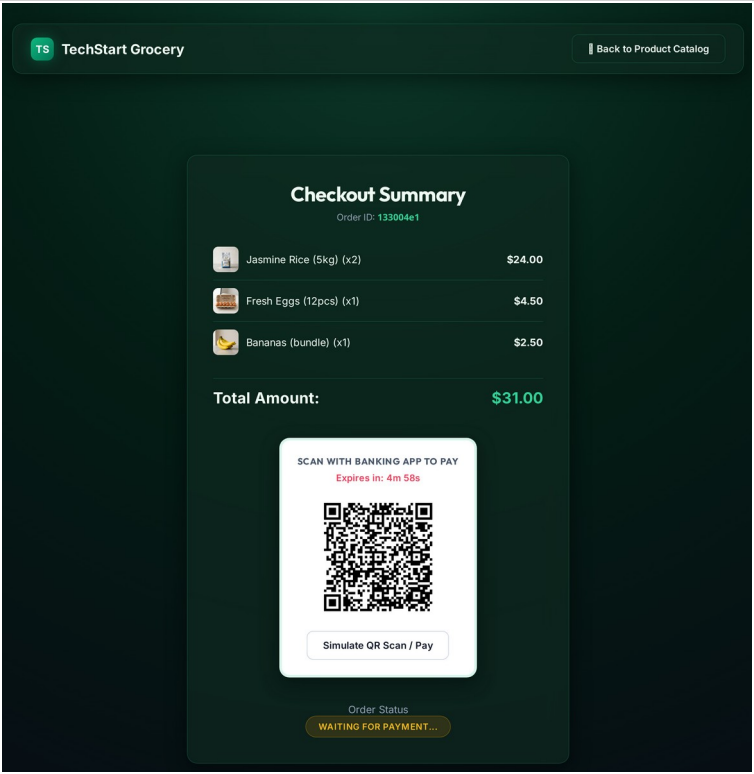
Redundant tunnels are used so the link to the core is not a single point of failure.

The application

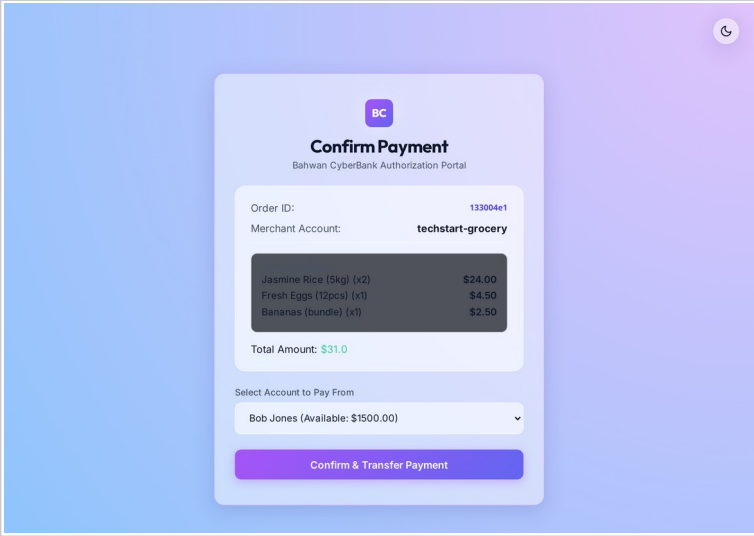
Merchant catalogue



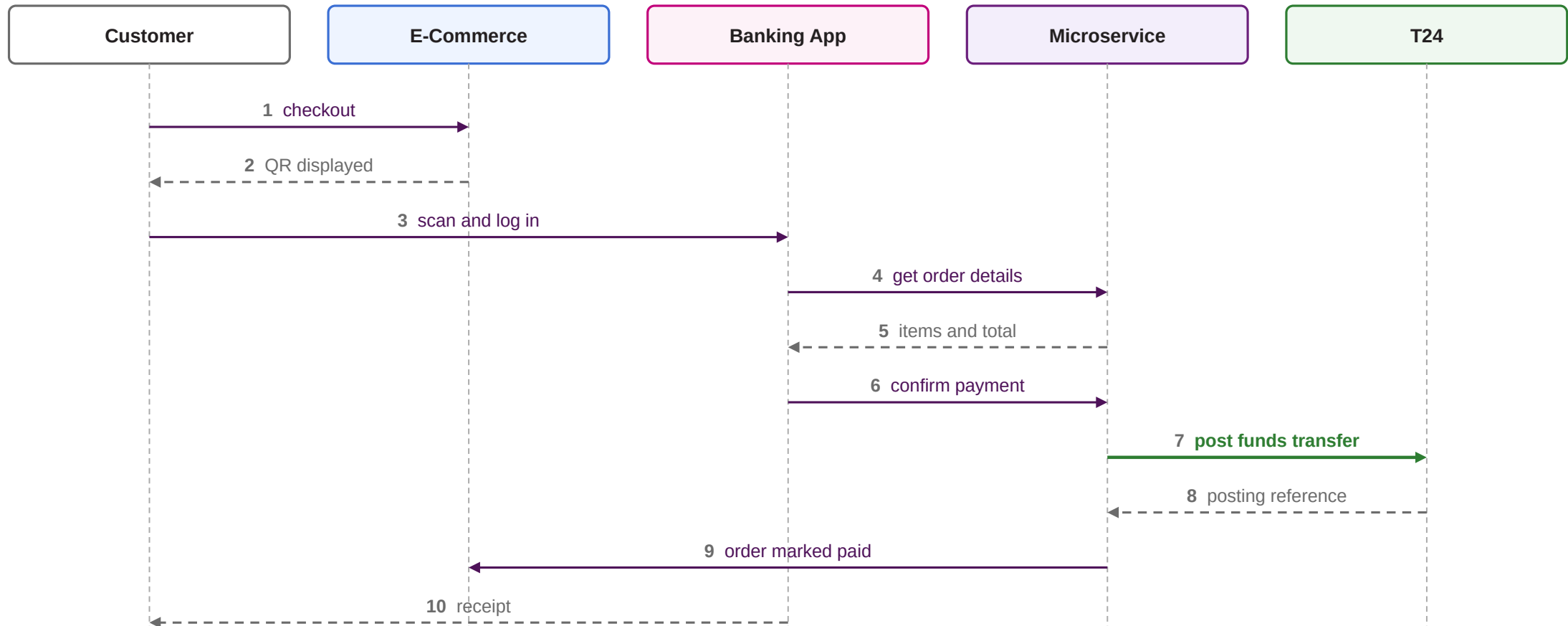
QR presented at checkout



Authorising in the bank app



Payment sequence



Step 7 is the only step that moves money, and it happens inside T24.

High availability

Application tier

Two or more replicas spread across availability zones, behind a health-probed ingress

Data tier

Zone-redundant managed database with automated backup and point-in-time restore

Deployment

Rolling updates gated on readiness probes, so a release does not interrupt service

Core banking link

Redundant site-to-site tunnels between Azure and the data centre

Availability targets

99.9%

service availability

5 min

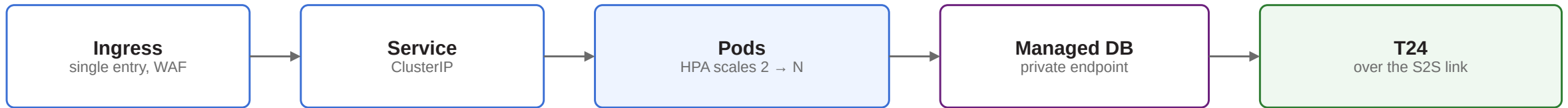
recovery point objective

1 hour

recovery time objective

Recovery objectives to be confirmed and signed off before build.

Scalability on Kubernetes



How it scales

- Horizontal Pod Autoscaler on CPU and requests per second
- Cluster autoscaler adds nodes when pods cannot be placed
- Pods are stateless, so any replica can serve any request

What makes that possible

- Session state held outside the pod
- Configuration supplied by environment and secret store
- Connection pooling in front of the database

Already in place

- Deployment, Service and Secret manifests written
- Liveness and readiness probes defined
- Health endpoints exposed by both services

Observability

Logs

Structured JSON from every container, shipped to a Log Analytics workspace with a defined retention period.

Metrics

Azure Monitor collects saturation, latency and error rate, surfaced on a workload dashboard.

Traces

OpenTelemetry with a correlation identifier carried across the channels, the microservice and the core.

Health and self-healing

- Liveness and readiness probes enforced by the platform
- Unhealthy pods removed from rotation automatically
- Alerts routed to an on-call rota

What we can answer

- Where a specific payment reached, and how long each hop took
- Whether an error came from the channel, the microservice or the core
- Whether the service met its availability target this month

BSP and regulatory alignment

Instrument	Requirement	How the design responds
Circular 1055	National QR code standard — QR Ph	EMVCo-format payload with a signed order reference
Circular 1198	Merchant payment acceptance	Merchant onboarding, settlement and dispute handling in the core
MORB Appendix 75	Information security risk management	TLS in transit, keys in Key Vault and HSM, immutable audit trail
Circular 1140	Technology outsourcing risk	Documented exit strategy, portable stack, continuity plan
RA 10173	Data privacy and residency	In-country archive copy; cross-border basis to be documented
RA 9160	Anti-money laundering reporting	Covered and suspicious transaction reporting from the core

Confirmed with Compliance and Legal

- The cross-border transfer basis and the licensing position are decided outside engineering

Standing position

- The core banking system remains the record for every posting and balance

Engineering mapping for discussion. Citations to be confirmed with Compliance and Legal.

Feedback and recommendations

Feedback

Recommendations

Action · owner
